Defining the problem

How do we turn a broad concern into a problem statement precise enough to analyse and act on?



Today

By the end of today, you will have

Completed the Problem Definition Canvas for your case and submitted the Case Approval Form.

- Write a problem statement that names population, place, institution, period and a measure of success.
- Bound the scope: what is in, and what is deliberately left out.
- Record competing framings and adopt one for a stated reason.
- Check your case against the feasibility tests.

Competencies: A2 · A3 · A4

Session plan

10 min	**Opening problem**
20 min	**Concept: precision and scope**
20 min	**Demonstration: canvas on Madhyapur**
30 min	**Activity: problem reframing lab**
30 min	**Own case: canvas and approval form**
10 min	**Debrief and exit task**

Opening problem

- **Government schools are failing. Who exactly, where, failing at what, compared with what, and who could change it?**

Rewrite the sentence so that a District Education Officer could act on it. You have three minutes.

The Problem Definition Canvas

Nine boxes. Fill them in order; each one narrows the problem.

Affected group

Who experiences the gap? Be specific: which households, workers or students.

Place

Which district, city, block or ward? One administrative unit is best.

Institution

Which office delivers or regulates the service in question?

Current condition

What is happening now, in observable terms, with a source and a year?

Desired condition

What should be true, by when, according to which rule, target or norm?

Decision-maker and decision

Which role can act, and what choice does it face?

Why it matters

Who is harmed, how badly, and is the harm unequal?

Out of scope

At least two things you will deliberately not analyse, with reasons.

Measure of success

How would anyone know the gap had narrowed?

5W1H turns a vague statement into a precise one

Question	Vague	Precise (Madhyapur)
Who?	Citizens	Households in residential wards
What?	Poor waste management	Waste handed over unseparated, contrary to the 2016 rules
Where?	Indian cities	Madhyapur Municipal Corporation
When?	Nowadays	On collection days in the current financial year
Why does it matter?	It is bad	Mixed waste goes to a near-full landfill; compostable and recyclable material is lost
How do we know?	Everyone says so	Ward inspection records and the corporation's returns (illustrative)

Madhyapur on the canvas

Affected group

Households in residential wards; the sanitation workers and waste pickers who handle mixed waste

Place

Madhyapur Municipal Corporation, all residential wards

Institution

The corporation's solid waste management department and its collection contractor

Current condition

On inspection days, most households hand over mixed waste (illustrative: about one in four separates)

Desired condition

Households hand over waste in the separate streams the 2016 rules require

Decision-maker and decision

Municipal Commissioner: which of three compliance strategies to fund next financial year

Why it matters

A near-full landfill; lost compost and recyclables; workers handling unsorted waste

Out of scope

Landfill remediation (a separate project); bulk and commercial generators (different rules and contracts)

Measure of success

Share of households handing over separated waste on inspection days, by ward

Narrowing a case: the ladder



1 Domain	2 Issue	3 Problem	4 Population	5 Place	6 Institution	7 Decision
Education	Foundational learning	Low reading levels at the end of Grade 3	Children in government primary schools	One named district	State department and district office	What the District Education Officer changes next year

A case that stops at the second rung is a field of study, not a problem. Keep climbing until a decision appears.

A thinker for your case

Amartya Sen

Development as Freedom, 1999

Sen argued that development should be judged by people's capabilities, the real freedoms they have to do and be what they value, not only by the resources they hold.

The question it puts to your case

Are you measuring access to a resource, or what people are actually able to do with it?

Answer it in one sentence in your workbook today.

One problem, three framings

1

The Commissioner

A compliance and cost problem: households are not following the rules, and mixed waste is expensive to dispose of.

2

Waste pickers

A livelihood and recognition problem: recyclables are lost in mixed waste, and their work is unrecognised and unsafe.

3

Residents near the landfill

A health and fairness problem: the city's mixed waste ends up next to their homes.

Your statement adopts one framing. Say which, and why it serves the decision. Record the others.

Is your case workable in four weeks? Five quick tests

1. **Specific**

   One problem, one group, one place, one institution.

2. **Evidence exists**

   You have already opened two public data sources and one published qualitative source.

3. **Someone can act**

   A named role has the authority and a real decision to make.

4. **Manageable**

   Analysable with published evidence in about fifteen hours of independent work.

5. **Safe**

   No primary data, no personal data, and analysable without taking a party-political position.

Activity: Problem reframing lab

Turn vague public concerns into precise, decision-focused statements, then test them against each other.

Scenario

Each pod receives four vague statements. Women do not work enough in India. Our cities have a waste crisis. Government schools are failing. Farmers do not get fair prices. Make each one usable by a real decision-maker.

What to do

1. Take one statement each. Run it through 5W1H in five minutes.
2. Rewrite it with population, place, institution and period.
3. Name the role that could act and the decision it faces.
4. Pass it to the pod sceptic, who must find one remaining vagueness.
5. Agree the pod's best rewrite and post it for a gallery walk.

Time 30 minutes

Groups Pods of four, one statement each

Output Four rewritten statements; one posted for the gallery

Your case: Worksheet 1.2

Problem Definition Canvas and Case Approval Form

1. Fill the nine boxes of the canvas for your chosen case.
2. Run your problem statement through 5W1H and fix any vague word.
3. Record two other framings and say which you adopt, and why.
4. Score your case on the five quick tests and fix any failure.
5. Complete the Case Approval Form and submit it today.

Hand in

Canvas and Case Approval Form by the end of Day 2. Provisional approval is returned by Day 3.

Quality check

The statement names group, place, institution and period.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. A precise statement names who, where, which institution, what gap, compared with what, and who could act.
2. Scope is a decision: say what you leave out and why.
3. Every problem can be framed several ways. Adopt one deliberately and record the others.
4. Define success by what changes for people, not only by what is spent or supplied.

Exit task Read your problem statement to your pod sceptic. Write down the one word they challenged.

Before the next session

Independent work

Revise your canvas after the gallery walk. Find the published qualitative or field-based source the Case Approval Form asks for, such as an audit report, social audit, committee report or field study, and note its two most relevant findings. About 45 minutes.

Read

Bardach and Patashnik, A Practical Guide for Policy Analysis, part I, step one: define the problem.

Next

Day 3: Structuring with issue trees

Tomorrow we break your problem into a structured set of questions with an issue tree, and you seal your first working answer.